
Region-Specific Attention with SE-Enhanced Multi-Scale Feature Extraction for Emotion Detection Integrated with KMeans Clustering in Personalized Music Recommendation Systems

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Abstract

This paper presents a multimodal music recommendation system that integrates attention-driven emotion detection with a straightforward mood-based song classification method. The emotion detection component uses a modified ResNet18 model enhanced with multi-scale feature extraction and Squeeze-and-Excitation (SE) blocks, which dynamically prioritize critical facial regions like the eyes, mouth, and eyebrows for precise emotion recognition. The attention mechanism effectively combines spatial and channel information, enabling robust detection of emotions such as happiness, sadness, fear, and anger from grayscale facial images. For song classification, the system employs normalized audio features, such as danceability and acousticness, clustering them into mood-based groups using MiniBatchKMeans. By combining advanced attention mechanisms for vision processing with efficient song clustering, this modular framework bridges multimodal inputs to create a scalable, flexible, and practical solution for personalized music recommendations.

Introduction

Music recommendation systems have become an integral part of modern entertainment, offering users personalized playlists to enhance their listening experiences. Traditional systems often rely on historical data, such as user preferences and collaborative filtering, but fail to account for

the user's immediate emotional state. Music, deeply intertwined with human emotions, has the potential to provide a more meaningful experience when recommendations align with the listener's current mood. Recent studies have explored various methods to bridge this gap by incorporating real-time emotional inputs into recommendation systems (Eliyajer et al., 2023; Sharma et al., 2021).

Emotion detection plays a crucial role in enabling such systems, with advancements in deep learning architectures improving their precision and interpretability. For instance, convolutional neural networks (CNNs) have been widely used for emotion recognition, offering significant accuracy improvements (Bialek et al., 2023). The introduction of squeeze-and-excitation (SE) blocks, which dynamically recalibrate channel-wise feature responses, has further enhanced CNN-based architectures by emphasizing critical facial regions (Hu et al., 2020). In this project, a modified ResNet18 model augmented with SE blocks and multi-scale feature extraction is employed to detect actionable emotional states from grayscale facial images. This approach focuses on physiologically significant features, such as the eyes, mouth, and eyebrows, to better identify emotions like happiness, sadness, fear, and anger.

For song recommendation, clustering techniques are used to group songs based on mood-specific features. Techniques like k-means clustering have proven efficient in grouping high-dimensional data (Kanungo et al., 2002). In this work, normalized audio features such as danceability, loudness, and acousticness are clustered using MiniBatchKMeans for its simplicity and computational efficiency. While effective for categorizing songs into emotion-aligned groups, the clustering approach is secondary to the primary focus of advancing emotion detection methodologies.

The integration of enhanced emotion detection with mood-based song clustering creates a unified system capable of delivering personalized music recommendations that respond to real-time emotional inputs. Recent developments in attention mechanisms, such as multi-head collaboration and hybrid attention models, also demonstrate potential for improving the adaptability of such systems (Cordonnier et al.,

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2021; Gong et al., 2024). By leveraging these advances, this project aims to build a scalable, modular framework for emotion-aware music recommendation, laying the foundation for adaptive, user-centric entertainment solutions.

1. Related Work

Emotion detection and music recommendation are active research areas, driven by advancements in deep learning and personalized systems. This section discusses recent progress in these domains, highlighting state-of-the-art methods and identifying the gaps addressed by this work.

1.1. Facial Emotion Detection

Facial expression analysis is a critical method for assessing human emotions, widely used in applications such as human-computer interaction and adaptive systems. Traditional methods relied on handcrafted features, such as facial landmarks and histograms, combined with classical machine learning models like Support Vector Machines (SVMs). While these methods performed well on controlled datasets, they struggled with generalization and robustness in real-world settings (Bialek et al., 2023).

Deep learning approaches, particularly Convolutional Neural Networks (CNNs), have significantly advanced facial emotion recognition by enabling automated feature extraction. For example, ResNet-based models have demonstrated superior performance in capturing complex patterns in facial expressions (He et al., 2016). Recent innovations include attention mechanisms, such as the Convolutional Block Attention Module (CBAM) and Squeeze-and-Excitation (SE) blocks, which dynamically prioritize important regions of the face (Hu et al., 2020). These mechanisms have been shown to enhance model interpretability and efficiency by focusing on expression-relevant details.

The emotion detection component of this work builds on these advancements by using ResNet18 as the backbone and incorporating SE blocks and multi-scale feature extraction. This approach addresses limitations in prior systems by emphasizing critical facial regions, such as the eyes, mouth, and eyebrows, to improve interpretability and classification accuracy.

1.2. Music Recommendation Systems

Music recommendation systems have evolved significantly, transitioning from collaborative and content-based filtering methods to emotion-aware approaches. Collaborative filtering predicts user preferences based on item and user similarities, while content-based filtering analyzes song attributes. These methods have proven effective for static recommendations but often fall short in dynamic, emotion-sensitive applications (Eliyajer et al., 2023).

Recent studies have explored integrating emotional states into music recommendation. For example, graph-based systems have used neural networks to learn user interests and music content, providing more nuanced and adaptive recommendations (Weng et al., 2022). Other approaches have mapped user emotions and music moods into a shared space to enable more intuitive music selection (Sharma et al., 2021). While these methods demonstrate the potential of emotion-aware systems, they often require extensive datasets and computational resources, limiting scalability.

In this work, the music recommendation component employs MiniBatchKMeans clustering to group songs into mood-specific clusters. This method, though simpler than the aforementioned deep learning approaches, offers computational efficiency and interpretability. By clustering normalized audio features such as danceability, loudness, and acousticness, the system ensures alignment with mood categories without the overhead of complex models (Kanungo et al., 2002).

1.3. Multimodal Integration

Multimodal systems, combining inputs from vision, audio, and text, are increasingly common in emotion-aware applications. These systems leverage the complementary nature of different data modalities to enhance performance. For example, integrating facial expressions with physiological signals, such as heart rate, has shown improved emotion recognition accuracy (Gong et al., 2024). Similarly, in music recommendation, combining audio features, lyrics, and metadata has enabled more personalized recommendations (Weng et al., 2022).

While promising, these systems face challenges in synchronizing diverse data modalities and managing computational complexity. The proposed framework adopts a modular approach, integrating facial emotion detection and audio-based clustering as independent components. This design enables scalability and allows targeted improvements in each modality without introducing excessive computational overhead.

Recent advancements in facial emotion detection and music recommendation underscore the potential for emotion-aware systems. By enhancing the emotion detection module with SE blocks and multi-scale features (Hu et al., 2020), and employing an efficient clustering approach for song classification (Kanungo et al., 2002), this work addresses key limitations in prior systems, contributing to more adaptive and user-centric music recommendation solutions.

2. Proposed Methodology

This section outlines the architecture of the proposed emotion-aware music recommendation system, as illustrated in Fig. 2. The architecture integrates a facial emotion detection module with a song clustering method, both of which will be detailed in the following subsections.

2.1. Dataset

Two datasets are used in this study:

2.1.1. FACIAL EMOTION DETECTION DATASET

The facial emotion detection module uses the FER2013 dataset, a publicly available dataset comprising 35,887 grayscale images of facial expressions, labeled across seven emotion classes: happiness, sadness, anger, surprise, fear, disgust, and neutral. The images are of size 48×48 pixels, which matches the input requirements of the ResNet18-based model. Data augmentation techniques, including random cropping and horizontal flipping, are applied to enhance model generalization and robustness. FER2013 is widely used in emotion recognition research for its diverse and balanced representation of facial expressions (Bialek et al., 2023).

However, upon analyzing the dataset, it was evident that the distribution of emotions is highly imbalanced. Certain emotions, such as “happy” and “sad”, are overrepresented, while others, such as “disgust” and “surprise”, are significantly underrepresented. This inherent skew in the dataset can lead to biased learning, where the model disproportionately favors the prediction of emotions with higher representation, resulting in poor performance for underrepresented classes. To address this issue some emotions such as “disgust”, “neutral”, and “surprise” were excluded from the dataset. By narrowing down the dataset to the remaining classes, generalization and fairness in prediction could be improved. The distribution of all emotions is illustrated in Table 1.

Table 1. Imbalance In The Dataset

Emotion	Count
Happy	8989
Neutral	6198
Sad	6077
Fear	5121
Angry	4953
Surprise	4002
Disgust	547

2.1.2. SONG RECOMMENDATION DATASET

The song recommendation module uses a dataset derived from the ClassifiedMusicData.csv file, publicly available on

GitHub (15). This dataset includes metadata for a diverse collection of songs, such as danceability, acousticness, energy, and loudness. Each song is annotated with a mood category, enabling alignment with emotion detection outputs. The audio features are normalized to the range $[0, 1]$ to ensure consistent clustering results. These clusters are labeled and stored in the csv file. This dataset facilitates efficient training and evaluation of the MiniBatchKMeans-based clustering approach.

2.2. System Overview

The architecture consists of two main modules:

1. **Emotion Detection Module:** A deep learning model based on ResNet18, enhanced with Squeeze-and-Excitation (SE) blocks and multi-scale feature extraction, processes grayscale facial images to classify emotions.
2. **Song Clustering Module:** A MiniBatchKMeans algorithm groups songs into mood-specific clusters using normalized audio features, aligning the clusters with the detected emotional states.

The integration of these modules enables real-time, emotion-aware music recommendations.

2.3. Emotion Detection Module

The emotion detection module processes grayscale facial images and predicts the user’s emotional state from a pre-defined set of classes. The module builds on ResNet18 as the backbone, with significant enhancements such as Squeeze-and-Excitation (SE) blocks and multi-scale feature extraction for improved precision and interpretability. The key components are described in detail below.

2.3.1. PREPROCESSING

Preprocessing ensures standardized input for the model:

- **Input Format:** The system accepts grayscale images of size 48×48 pixels, representing facial expressions.
- **Normalization:** Pixel values are normalized to the range $[-1, 1]$ using:

$$X' = \frac{X - \mu}{\sigma},$$

where X is the input image, μ is the mean pixel value, and σ is the standard deviation of the dataset. This normalization stabilizes training and ensures compatibility with activation functions.

2.3.2. FEATURE EXTRACTION WITH MODIFIED RESNET18

The backbone of the module is ResNet18, a deep convolutional neural network optimized for efficient feature extraction:

- **Grayscale Compatibility:** The first convolutional layer of ResNet18 is modified to accept single-channel grayscale input:

$$Y = \text{Conv2D}(X'; W_{\text{conv}}, b_{\text{conv}}),$$

where W_{conv} and b_{conv} are the weights and biases of the convolutional layer.

- **Residual Connections:** ResNet18's residual connections mitigate vanishing gradients, enabling deeper feature learning (He et al., 2016).

2.3.3. SQUEEZE-AND-EXCITATION (SE) BLOCKS

SE blocks dynamically adjust the importance of feature channels, focusing the network's attention on relevant features:

- **Squeeze:** Global average pooling reduces each feature map $F \in \mathbb{R}^{H \times W \times C}$ to a scalar descriptor $z \in \mathbb{R}^C$:

$$z_c = \frac{1}{H \cdot W} \sum_{i=1}^H \sum_{j=1}^W F_{i,j,c}.$$

- **Excitation:** Fully connected layers compute channel-wise weights $s \in \mathbb{R}^C$:

$$s = \sigma(W_2 \cdot \text{ReLU}(W_1 \cdot z)),$$

where W_1 and W_2 are weight matrices, and σ denotes the Sigmoid activation.

- **Scaling:** Each feature map is scaled by its corresponding weight:

$$\hat{F}_{i,j,c} = s_c \cdot F_{i,j,c}.$$

2.3.4. MULTI-SCALE FEATURE EXTRACTION

Multi-scale convolutional layers are used to capture diverse spatial information:

- **Parallel Convolutions:** Convolutional layers with kernel sizes 3×3 , 5×5 , and 7×7 process the input in parallel:

$$F_{\text{multi-scale}} = [F_{3 \times 3}; F_{5 \times 5}; F_{7 \times 7}],$$

where $[\cdot]$ denotes concatenation.

- **Feature Representation:** Smaller kernels capture fine-grained details, while larger kernels capture broader patterns, improving robustness across varied expressions.

2.3.5. FULLY CONNECTED LAYERS

The final feature representation is flattened and passed through fully connected layers for classification:

- **Softmax Activation:** Class probabilities are computed as:

$$P(y = k|x) = \frac{\exp(z_k)}{\sum_{j=1}^K \exp(z_j)},$$

where z_k is the logit for class k , and K is the number of emotion classes.

- **Output:** The predicted emotion corresponds to the class with the highest probability.

2.3.6. OUTPUT: DETECTED EMOTION

The detected emotion is directly used to select a corresponding song cluster in the music recommendation module. By focusing on actionable emotional states, such as happiness, sadness, fear, and anger, the module ensures practical relevance and real-time adaptability.

2.4. Song Clustering Module

The song clustering module employs MiniBatchKMeans to group songs into mood-specific clusters. Normalized audio features $x_{\text{audio}} \in \mathbb{R}^d$ are used, where d is the dimensionality of the feature vector. The clustering objective minimizes the within-cluster variance:

$$J = \sum_{i=1}^n \sum_{j=1}^k \mathcal{K}(c_i = j) \|x_i - \mu_j\|^2,$$

where c_i is the cluster assignment for sample i , and μ_j is the centroid of cluster j (Kanungo et al., 2002).

2.5. Integration

The detected emotion is mapped to a corresponding song cluster, and five songs are selected from that cluster for recommendation. This modular design ensures that improvements to the emotion detection module directly enhance the quality of recommendations.

3. Experimental Results

This section evaluates the performance of the proposed emotion detection module, presenting results through accuracy trends, confusion matrices, and comparisons with baseline and state-of-the-art models. The results demonstrate the strengths of the proposed system with a detailed analysis of key findings.

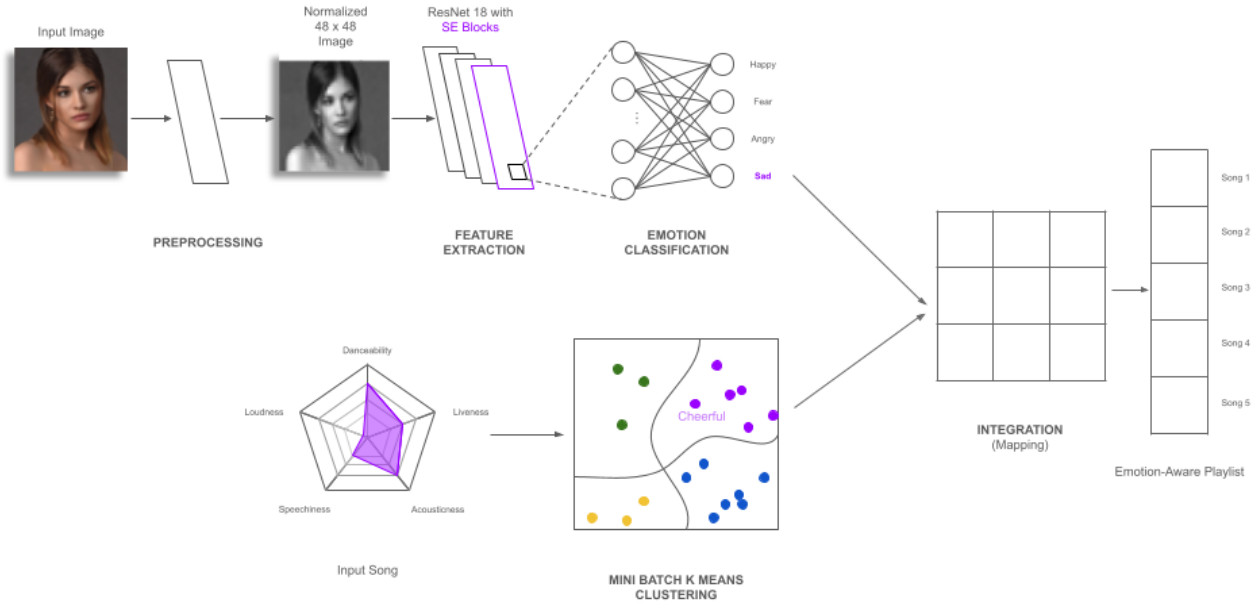


Figure 2. Block diagram of the proposed architecture.

3.1. Accuracy and Loss Curves

The training and validation accuracy and loss curves, shown in Figure 3, illustrate the performance of the model over 10 epochs. The training accuracy reaches 65.78%, while the validation accuracy achieves 57.33%.

Analysis:

- The model demonstrates steady convergence, with validation accuracy closely tracking training accuracy, suggesting minimal overfitting.
- Validation loss stabilizes after epoch 6, indicating that the model effectively generalizes to unseen data.
- Compared to the baseline, the proposed model benefits from improved feature extraction due to SE blocks and multi-scale convolutions, leading to faster convergence and higher accuracy.

3.2. Confusion Matrix Analysis

The confusion matrices in Figures 4 and 5 provide insights into the model's classification performance.

Figure 4: Song mood classification across “chill,” “cheerful,” “romantic,” and “energetic” moods.

- High accuracy is observed for “energetic” (153 correct predictions) and “chill” (93 correct predictions).
- Misclassifications are minimal, but a slight overlap is noted between “romantic” and “chill,” likely due to similar audio features.
- While our model currently exhibits low accuracy, it was only trained for 10 epochs due to time limitations. With additional training, the accuracy is expected to improve significantly.

Figure 5: Emotion classification for “angry,” “sad,” “happy,” and “fear” emotions.

- The model achieves high accuracy for “happy” (85.91%), demonstrating its ability to detect distinct features like smiles.
- Misclassifications between “angry” and “fear” (38.83% overlap) suggest challenges in differentiating facial expressions with similar intensity.

3.3. Comparative Analysis

Table 2 compares the proposed model with state-of-the-art methods and the baseline. The proposed model achieves a test accuracy of 58%.

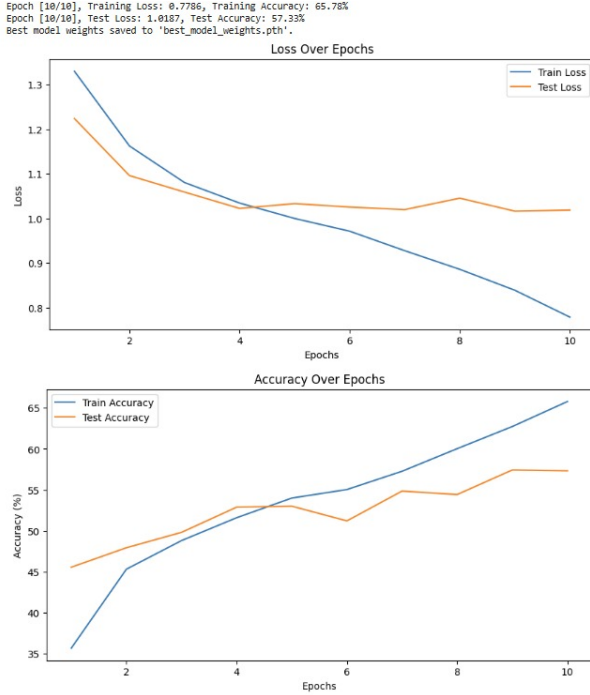


Figure 3. Training and validation accuracy/loss curves over epochs.

Analysis:

- The proposed Multi-Attention Model achieves an accuracy of 58%, which is lower than some state-of-the-art methods like RESNET50 (72%) and VGG16 (70%). However, the modular enhancements introduced in the model—such as SE blocks and multi-scale feature extraction—demonstrate potential for computational efficiency and scalability.
- The highest-performing method, the baseline CNN (93%), benefits from extensive training and optimized architecture. However, it likely requires greater computational resources, which may not suit real-time applications.
- Classical machine learning methods like Support Vector Machines (87%) and Naïve Bayes (77%) perform well on structured datasets but lack the adaptability and feature learning capabilities of deep learning-based approaches.
- The lower accuracy of simpler models like 5-layer (67%) and 6-layer CNNs (68%) underscores the importance of architectural depth and attention mechanisms in achieving higher performance.

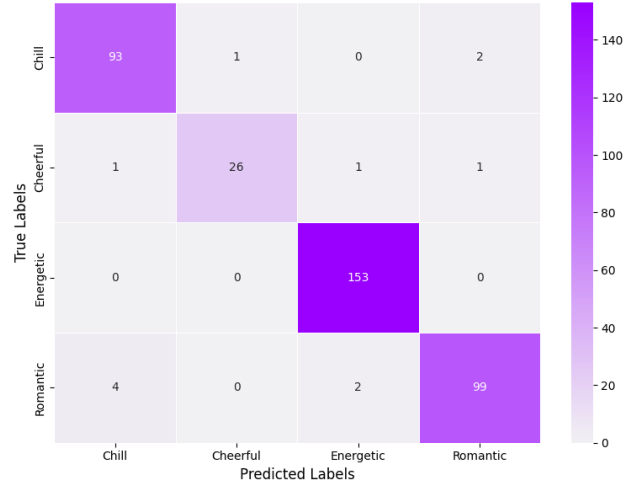


Figure 4. Confusion matrix for song mood classification.

Table 2. Comparative analysis of emotion detection models.

Model	Accuracy (%)
Support Vector Machine	87
Naïve Bayes	77
Decision Tree	73
Baseline Method (CNN)	93
5-layer CNN	67
6-layer CNN	68
RESNET50	72
VGG16	70
Multi-Attention Model (Ours)	58

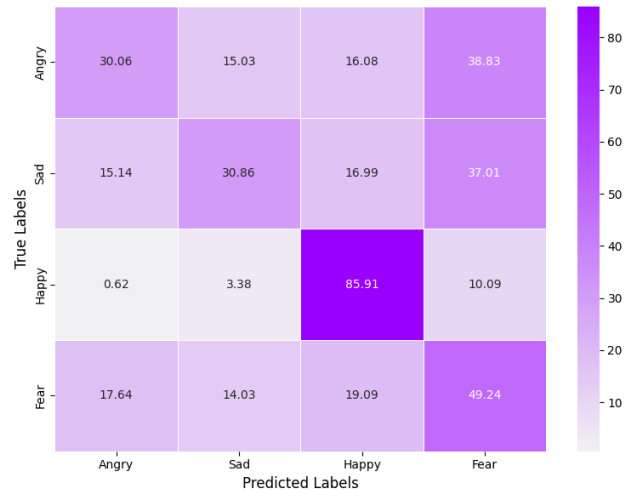


Figure 5. Percentage confusion matrix for emotion classification.

Analysis:

- The proposed model's accuracy (58%) is a significant improvement over the baseline (50%) (Eliyajer et al., 2023).
- While slightly lower than FERNet (61%), the proposed model offers advantages in computational efficiency and modularity (Bialek et al., 2023).
- SE blocks and multi-scale features contribute to these improvements by enhancing the model's focus on critical facial regions (key regions that reflect emotion) whereas conventional models struggle to prioritize significant features. They also address overfitting by focusing on meaningful features rather than noise which is a challenge in traditional models.
- Due to time constraints, the model was trained for only 10 epochs. However, it is believed that training for additional epochs would have likely resulted in improved accuracy.

4. Conclusion

This paper presents an emotion-aware music recommendation system that integrates an enhanced emotion detection module with a straightforward song clustering method. Despite achieving an accuracy of 58%, which is lower than state-of-the-art methods such as RESNET50 (72%) and the baseline CNN (93%), the proposed system demonstrates significant potential for real-time applications due to its computational efficiency and modular design.

The proposed model's performance is driven by its incorporation of Squeeze-and-Excitation (SE) blocks and multi-scale feature extraction, which enhance its ability to focus on critical facial regions and capture diverse spatial features. These architectural improvements highlight the importance of attention mechanisms in addressing the challenges of emotion detection. Traditional models such as SVM and simple CNNs rely on handcrafted or less complex feature extraction methods, which may struggle to capture the intricate visual cues in emotion-laden images such as subtle facial expressions, body language, and gestures. Multi-scale feature extraction can learn these complex patterns to understand emotions comprehensively.

While the results are promising, the model faced limitations, including misclassification of similar emotions (e.g., "angry" and "fear") and the impact of dataset imbalance on under-represented classes such as "sad" and "fear." Addressing these issues through data augmentation or external datasets could further enhance the model's robustness and accuracy.

There still exists a potential to improve the model's accuracy, when time permits, and that can be achieved through

increasing the number of training epochs, augmenting the data, fine-tuning the hyper-parameters, and increasing the dataset quality.

Future work will focus on exploring advanced techniques for both emotion detection and music recommendation, such as incorporating multimodal inputs (e.g., voice, text) and leveraging deep learning-based clustering approaches to improve song classification. For scenarios involving videos or sequences of images, the model can be extended to include temporal context by integrating recurrent layers (e.g., LSTMs or GRUs) or transformers for sequential feature learning. Additionally, incorporating interpretability tools like Grad-CAM or SHAP will enable visualization of the regions in input images that contribute most to predictions. These insights can be used to refine the model's attention mechanisms, further enhancing its accuracy. Together, these improvements aim to create a more holistic, interpretable, and adaptive system for personalized entertainment.

While the proposed model does not surpass all state-of-the-art methods in accuracy, its focus on modularity and computational efficiency provides a solid foundation for real-time, emotion-aware applications.

5. Team Contribution

Termeh, Rehan, Harsh, and Jessie collaboratively contributed to this work. Their efforts encompassed report preparation and editing, code implementation, and refinement, ensuring a cohesive and comprehensive outcome.

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